
Replacing the Bispectrum: A Compact Learned Invariant Descriptor Preserves Quadratic-SNAP Accuracy with Less Data

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Abstract

The quadratic Spectral Neighbor Analysis Potential (SNAP) couples a hand-engineered descriptor— $K=55$ bispectrum components of the local neighbor density—to a linear-plus-quadratic regression head fit by least squares. We ask what changes when the descriptor itself is learned from data while everything else in the SNAP pipeline is held fixed. We introduce SLIM (Spectrally-Learned Invariant Moments), a compact descriptor that contracts learned radial channels against spherical-harmonic moments into per-degree power spectra and compresses them with a small network into $K'=32$ features; rotation, translation, and permutation invariance hold by construction, the isolated-atom energy is exactly zero, and forces are analytic gradients. The quadratic head on SLIM has 560 least-squares coefficients versus 1595 for quadratic SNAP. On the tantalum training set of Wood and Thompson [J. Chem. Phys. 148, 241721 (2018)], SLIM-SNAP reaches $\delta E = 0.00281$ eV/atom and $\delta F = 0.00987$ eV/Å at full data against $\delta E = 0.00825$ eV/atom and $\delta F = 0.01502$ eV/Å for our reimplemented quadratic SNAP (paper-as-reported: 0.00898 eV/atom). Under the paper’s cross-validation protocol (5% increments, ten draws per fraction), the smaller head eliminates quadratic SNAP’s low-data overfitting catastrophe outright, and on the discriminating low-symmetry subset (GSF, Liquid, Surface) SLIM-SNAP reaches the baseline’s full-data accuracy with 60% of the training data (baseline: 65%), improving to 25% for a $K'=16$ variant whose head has $10\times$ fewer coefficients than quadratic SNAP’s; at the cross-validation endpoint SLIM-SNAP’s held-out subset error is $7.7\times$ lower. The same architecture transfers to Mo, Nb, and Re (HCP), where quadratic SNAP is capacity-limited rather than data-limited and SLIM lowers the held-out error floor at essentially every training fraction, reaching the baseline’s full-data accuracy with 5–10% of the data. An invariance audit confirms drift at floating-point precision, and a lab-frame ablation quantifies what learning the invariance from rotation-augmented data would cost instead. We discuss mechanisms, failure cases, and inference cost.

1 Introduction

Machine-learned interatomic potentials (MLIPs) of the density-projection family—SNAP [1], GAP/SOAP [2, 3], MTP [4], ACE [5]—factor the map from an atomic environment to a per-atom energy into two stages: a hand-engineered *descriptor* that projects the neighbor density onto a fixed invariant basis, and a *regression head* fit to reference energies and forces. The descriptor encodes rotation, translation, and permutation invariance by construction, so the regression inherits the symmetries of physics for free; this is widely credited for the data efficiency of the family, since no training configuration is spent teaching the model that energy does not depend on orientation. Wood and Thompson’s quadratic SNAP [6] is a sharp instance of this design: $K=55$ bispectrum

components $\mathbf{B}^i$ of the neighbor density feed the energy form

$$E^i = \beta \cdot \mathbf{B}^i + \frac{1}{2} (\mathbf{B}^i)^\top \boldsymbol{\alpha} \mathbf{B}^i, \quad (1)$$

whose $K + K(K+1)/2 = 1595$ coefficients are obtained by linear least squares. On a tantalum training set of 363 DFT configurations the quadratic terms buy a $3.4\times$ reduction in energy error over linear SNAP at the same descriptor count—at the price, documented in their cross-validation analysis, of catastrophic overfitting when less than $\sim 30\%$ of the training data is used, because the coefficient count then exceeds the number of training rows.

This paper asks a question that the quadratic-SNAP design makes unusually crisp: *what changes if the descriptor is learned from data rather than hand-engineered, while every other part of the pipeline is held fixed?* We keep the energy form of Eq. 1, the least-squares solver, the zero-energy constraint for an isolated atom, analytic-gradient forces, the training data, and the published cross-validation protocol, and we replace only $\mathbf{B}^i$: the bispectrum gives way to a learned map $\mathbf{B}_\theta^i$ with $K'=32$ outputs. Because the head is linear in its coefficients, the substitution isolates the descriptor’s contribution cleanly—any change in accuracy, data appetite, or symmetry behaviour is attributable to the representation, not to a more powerful regressor. An end-to-end neural energy model would conflate the two; we deliberately do not build one.

Our descriptor, SLIM (*Spectrally-Learned Invariant Moments*), contracts learned radial channels against real spherical-harmonic moments of the neighbor density into per-degree power spectra and compresses those invariants with a small multilayer perceptron. Three properties are architectural: rotation, translation, and permutation invariance hold exactly (the audit in Sec. 5.4 measures drift at floating-point rounding, 10^{-13} – 10^{-16} relative); the isolated-atom output is subtracted so the isolated-atom energy is exactly zero, mirroring SNAP’s `bzerooflag`; and the map is smooth, so forces are analytic gradients. The learned part is deliberately compact: with $K'=32$ the quadratic head has 560 coefficients, $2.8\times$ fewer than quadratic SNAP’s 1595, while the descriptor itself carries $\sim 253\text{k}$ trainable parameters that are fit, per training subset, by gradient descent on the same energy and force rows the head sees—never on held-out data.

Holding the head fixed turns the descriptor question into a set of measurable claims. First, *accuracy*: at full training data on tantalum, SLIM-SNAP must match a faithfully reimplemented quadratic SNAP on both energy and force MAE to be interesting at all. Second, *data efficiency*: the paper’s own cross-validation protocol (training fractions swept in 5% increments with ten stratified draws per fraction) exposes how each descriptor’s error decays with data, and the smaller head moves the overfitting cliff. Third, *generalization breadth*: the same architecture must fit other refractory metals—we use the curated Mo, Nb, and Re datasets of Andolina and Saidi [7], with Re’s HCP structure serving as the structural stress test. Fourth, *invariance*: a learned descriptor’s symmetry properties must be demonstrated, not asserted, and their value quantified; we do both, including a lab-frame ablation that learns rotation invariance from data augmentation instead of encoding it.

Contributions. (i) SLIM, a compact learned invariant descriptor that drops into the SNAP pipeline unchanged, with a training procedure—joint gradient descent on descriptor and a surrogate head, followed by an exact least-squares head re-solve—that makes the deployed model identical in form and solver to quadratic SNAP. (ii) A complete reproduction of the Wood–Thompson cross-validation protocol for both the reimplemented baseline and SLIM-SNAP on Ta, Mo, Nb, and Re: over 1000 descriptor trainings and least-squares fits, reported per group, per fraction, and per draw. (iii) A mechanism study separating three ingredients—head size, architectural prior, and learning—using coefficient-count scaling, a random-feature control, and lab-frame ablations with and without rotation augmentation. (iv) An honest accounting of where the learned descriptor does *not* help, including groups where quadratic SNAP remains ahead and the regime where the transfer datasets make both models capacity- rather than data-limited.

2 Background: quadratic SNAP and its data appetite

SNAP [1] writes the total energy of a configuration as $E = E_{\text{ref}} + \sum_i E^i$, where E_{ref} is a ZBL screened-Coulomb reference pair potential [8] that handles short-range repulsion, and E^i depends on the neighbors of atom i within a cutoff r_c through the bispectrum components $\mathbf{B}^i$ of the neighbor density mapped onto the 3-sphere. The components are shifted by their isolated-atom values (`bzerooflag=1`), and the constant term is constrained to zero, so an isolated atom has exactly zero

SNAP energy. Forces are analytic gradients. In linear SNAP the head is $\beta \cdot \mathbf{B}^i$; quadratic SNAP (Eq. 1) adds all pairwise products, which Ref. [6] motivates as an EAM-like embedding-energy correction and shows to raise the effective body-order from four to seven.

Two empirical facts from Ref. [6] frame our study. At full training data, quadratic SNAP at $J_{\max}=4$ ($K=55$, $r_c=5.594$ Å) reaches $\delta E = 0.00898$ eV/atom on the 363-config tantalum set under unweighted least squares (their Table SII), a $3.4\times$ improvement over linear SNAP at matched K (Table SI: 0.03041 eV/atom at $r_c=5.672$ Å). And under cross-validation with 5%-increment training fractions and ten random draws per fraction, the quadratic form’s held-out error explodes below $\sim 30\%$ included data, where the 1595 coefficients exceed the training rows; disordered groups (generalized stacking faults, liquid, surfaces) converge much more slowly than high-symmetry crystal groups. The bispectrum is fixed *a priori*, so the only lever against this data appetite is the head—and shrinking $J_{\max}$ costs accuracy at full data. A learned descriptor offers a different trade: let the representation adapt to the material so that a *smaller* invariant vector, and therefore a much smaller quadratic head, suffices.

3 SLIM: a learned descriptor under the SNAP energy form

3.1 Architecture

SLIM maps the environment of atom i —relative neighbor positions $\mathbf{x}_{ij}$ within a cutoff r_c , including periodic images—to $\mathbf{B}_\theta^i \in \mathbb{R}^{K'}$ in four stages (all defaults given here are the settings used for every material):

(1) *Learned radial channels.* Each pair distance r is expanded in $N_b=12$ Chebyshev polynomials $T_n(2r/r_c - 1)$ multiplied by the SNAP cosine cutoff envelope $f_c(r) = \frac{1}{2}[\cos(\pi r/r_c) + 1]$, and mixed by a learned matrix $W \in \mathbb{R}^{C \times N_b}$ into $C=16$ radial filters $g_c(r) = \sum_n W_{cn} T_n f_c$. The filters vanish smoothly at r_c , so energies and forces are smooth when neighbors cross the cutoff. (An MLP parameterization of the filters was tried and retired: at matched budget it was no better than linear mixing; see the ablation grid in Sec. 6.)

(2) *Equivariant moments.* With real orthonormal spherical harmonics Y_{lm} up to $L=4$,

$$A_{clm}^i = \sum_{j \in \mathcal{N}(i)} g_c(r_{ij}) Y_{lm}(\hat{\mathbf{x}}_{ij}). \quad (2)$$

The sum over neighbors gives permutation invariance; relative coordinates give translation invariance; under a rotation the A_{cl}^i transform within each degree l by the orthogonal Wigner matrix D^l .

(3) *Invariant contraction.* Per-degree power spectra across ordered channel pairs,

$$P_{cc'l}^i = \sum_{m=-l}^l A_{clm}^i A_{c'lm}^i, \quad c \leq c', \quad (3)$$

are exactly rotation-invariant because D^l is orthogonal. Together with the first-order channel sums $t_c^i = A_{c00}^i$ (two-body information that pure quadratic contractions lose), this yields $z^i \in \mathbb{R}^{n_{\text{inv}}}$ with $n_{\text{inv}} = C + \binom{C+1}{2}(L+1) = 696$ invariants.

(4) *Learned compression.* A two-layer SiLU MLP ($696 \rightarrow 256 \rightarrow 256 \rightarrow K'$) maps standardized invariants to the descriptor. Standardization constants are frozen at initialization from the current training subset (a label-free statistic). The isolated-atom output is subtracted, $\mathbf{B}_\theta^i = \text{MLP}(z^i) - \text{MLP}(z_0)$ with z_0 the zero-neighbor invariant vector, so $\mathbf{B}_\theta(\text{isolated}) = 0$ exactly—the `bzero lag=1` analog—and a per-feature RMS scale (fitted on the training subset, absorbed into the head) keeps the least-squares system well-conditioned. Total: 252,640 trainable parameters, $K'=32$ outputs, and a quadratic head with $K' + K'(K'+1)/2 = 560$ least-squares coefficients.

Invariance is therefore architectural: rotation invariance enters through Eq. 3, not through data. Sec. 6 deploys a deliberately weakened variant (“lab-frame”) that feeds the uncontracted moments A_{clm}^i directly to the MLP, with and without rotation-augmented training, to price this design choice.

3.2 Training procedure

For a given training subset (an *included* set of configurations from one cross-validation draw), SLIM is trained from scratch:

1. **Joint gradient phase.** θ and a surrogate head (β, α) are optimized by Adam (cosine-annealed learning rate $3 \times 10^{-3} \rightarrow 10^{-5}$, gradient-norm clip 1.0, minibatches of 16 configurations) on the same rows the least-squares solver sees: per-atom energy residuals $(E_{\text{pred}} - E_{\text{DFT}} + E_{\text{ZBL}})/N$ and force-component residuals, with forces obtained by back-propagation through Eq. 2–3. Minibatch configurations are sampled with probability proportional to their atom count, which makes the sampled row distribution match the least-squares row measure (a configuration contributes $3N$ force rows); this single choice, not extra capacity, is what closed the last $\sim 40\%$ of the force-accuracy gap during development (uniform sampling starves the force-rich 100-atom liquid cells; see the attempts log). Energy rows are upweighted by $\gamma_E=4$ during this phase only (they are outnumbered 35:1 by force rows); the step budget is $s(f) = 6000 + \lceil f \cdot (60,000 - 6000) \rceil$ at included fraction f , fixed a priori for all draws, materials, and ablations.
2. **Exact head re-solve.** θ is then frozen and (β, α) is discarded and re-solved by the *identical* least-squares path as the baseline: the SNAP design matrix (energy rows and analytic force rows, assembled from $\mathbf{B}_\theta$ and its position Jacobians in float64) is solved with SVD-based `scipy.linalg.lstsq` at condition cutoff 10^{-13} —byte-for-byte the solver settings of FitSNAP’s SVD solver—on the included rows only. The deployed model is Eq. 1 with these coefficients.

No held-out configuration influences either stage: descriptor training, standardization, feature scaling, and the head solve all see the included portion of the current draw only, and hyperparameters were selected once on full-data *training* metrics (Sec. 4.3) before any sweep was run. Training one draw takes ~ 21 minutes at full data on one GPU; the sweeps in this paper comprise over 1000 such trainings.

3.3 Design rationale

Three choices deserve explicit motivation. *Compactness* ($K'=32$, $K=55$): the head’s coefficient count sets the position of the low-data overfitting cliff; a compact descriptor attacks the data-appetite problem at its source, and the learned nonlinear compression is what must make compactness affordable. *Power spectrum rather than bispectrum*: the per-degree power spectrum is the cheapest complete-enough invariant set when followed by a nonlinear network; known counterexamples to power-spectrum completeness [9] are measure-zero configurations that did not manifest in any of our experiments (the MLP mixes $C=16$ channels across five degrees, which breaks the single-channel degeneracies in practice). *Frozen-descriptor least-squares deployment*: keeping the deployed head an exact LSQ solution preserves everything the SNAP ecosystem values—fast refits under new weights, linear uncertainty analysis, and a paired comparison in which the only changed ingredient is the descriptor.

4 Experimental protocol

4.1 Data, baselines, and reference potential

Tantalum. The 363 DFT configurations (12 groups; 363 energy rows and 12,672 force rows) distributed with FitSNAP [10] as `Ta_Linear_JCP2014/JSON`, identical to the set used by Ref. [6]. We use them exactly as shipped; no augmentation, no additional DFT.

Transfer materials. The curated single-element DNP datasets of Andolina and Saidi [7] for Mo (3727 configurations), Nb (3246), and Re (5420, HCP), converted frame-by-frame from DeePMD format to per-configuration FitSNAP JSONs. Cross-validation groups follow the source tree: adaptive-learning iteration (with the initial iteration split by MD temperature) $\times$ configuration category (elastic-distorted MD, vacancies, interstitials, plain MD), giving 12 groups for Mo and Re and 16 for Nb. A constant per-atom energy shift (the dataset mean, one number per material, applied identically to both methods) removes the raw VASP atomic reference. Per-material cutoffs are set by a fixed physical rule stated in

advance of any fitting—the tantalum baseline cutoff scaled by the ratio of nearest-neighbor distances of the equilibrium structures, $r_c^X = 5.594 \times d_{nn}^X / d_{nn}^{\text{Ta}}$, giving 5.33 Å (Mo), 5.59 Å (Nb), and 5.36 Å (Re)—and used identically by the quadratic-SNAP baseline and SLIM on each material, so no cutoff tuning enters the comparison anywhere.

One descriptor *architecture* is used for every material, with per-material weights θ trained per draw (the preferred design under the study’s own framing: a uniform construction makes the cross-material claim about the architecture, not about per-material engineering). As a zero-extra-cost probe of descriptor transferability we also evaluate a *frozen-Ta* variant on the transfer materials: θ from the full-data tantalum model is reused unchanged—the learned radial filters live on the dimensionless coordinate r/r_c , so transferring them to another element rescales the shells by that element’s cutoff—and only the (β, α) head is re-solved per draw. This variant is trivially leakage-clean for the transfer sweeps (θ never saw any Mo/Nb/Re configuration) and separates what the descriptor learned about *geometry* from what it learned about the element.

Two baselines. Every comparison reports (i) *paper-as-reported* quadratic SNAP, cited from Wood–Thompson Tables SI/SII with the $J_{\text{max}}=4$ row ($\delta E = 0.00898$ eV/atom, $r_c = 5.594$ Å) as the headline anchor, and (ii) *reimplemented* quadratic SNAP: our own FitSNAP 3.1.1 run at $J_{\text{max}}=4$ (twojmax= 8, $K=55$, $r_c=5.594$ Å, $r_{\text{fac0}}=0.99363$, bzero flag=1, quadratic flag=1) on the shipped dataset under the paper’s Sec.-V protocol—unweighted least squares on energy and force rows—which is the paired-comparison reference for SLIM. The reference potential is ZBL (4.0–4.8 Å switching) subtracted from labels by LAMMPS inside FitSNAP for both methods; our standalone reimplement of ZBL agrees with LAMMPS to 10^{-13} (machine precision) in energy and forces.

Reimplementation gap. Our full-data quadratic SNAP gives $\delta E = 0.00825$ eV/atom and $\delta F = 0.01502$ eV/Å, 8% below the paper’s 0.00898. Exact reproduction is impossible on the distributed dataset: the paper reports 15,297 force rows (its Liquid group alone contributes 4203) while the shipped set yields 12,672 (Liquid: 900), and the linear-SNAP anchor is only recovered to within 21% under a total-energy row convention (0.0367 vs. 0.03041) while the quadratic anchor prefers the per-atom convention we adopt. We treat the 8% gap as the irreducible provenance uncertainty of the anchor; since both methods are evaluated against the *same* reimplemented baseline on the *same* rows, the paired comparison is unaffected. Appendix A tabulates the protocol-reconstruction evidence.

4.2 Cross-validation protocol

We reproduce the Figure-5 protocol of Ref. [6] exactly: training fractions $f \in \{0.05, 0.10, \dots, 0.95\}$, ten random draws per fraction, stratified within each group (included count $\text{round}(f n_g)$, clipped to $[1, n_g]$), a configuration’s energy and force rows always kept together, and the full-data point ($f=1$) reported as the accuracy-parity headline. Draws are generated once from a seeded generator and *shared* between the baseline and every SLIM variant, so all comparisons are paired draw-by-draw. Both methods are solved by the same least-squares settings on the same cached label vector; energy MAE (δE , eV/atom) and force MAE (δF , eV/Å) are recorded on the included and excluded portions for every (material, fraction, draw, group) cell—the complete per-cell table accompanies the logs. Proportional rounding makes some small-group held-out cells structurally empty at high fractions (e.g. the three-configuration Liquid group is fully included for $f \geq 0.85$); per-group curves simply omit those points, and subset aggregates average over the groups that have held-out members at that fraction, identically for both methods. For the large transfer systems the identical weighted system is solved by QR least squares on GPU, validated against the CPU SVD path to 2×10^{-8} relative (Appendix A); rank-deficient cells (which occur only for the baseline’s 1595-column system at small Ta fractions) always use the SVD minimum-norm path, matching FitSNAP.

Deviations and disclosures, consolidated. For auditability we collect every choice that goes beyond the strictly-reproduced protocol in one place. (i) The paper’s Table-SI/SII row convention cannot be reconstructed exactly (Appendix A); we use per-atom energy rows for both methods. (ii) Stress rows are excluded from all fits (the Table-SI/SII protocol fits energies and forces; the shipped example’s virial weight is 10^{-12}). (iii) Transfer-material labels are shifted by one constant per material (the dataset-mean per-atom energy), identically for both methods. (iv) Transfer cutoffs follow the stated nearest-neighbor scaling rule; no tuning. (v) SLIM’s per-draw training budget follows the pre-stated ramp $s(f)$; the lab-frame ablations run on a disclosed reduced grid of seven fractions $\times$ ten draws. (vi) Proportional rounding leaves some small-group held-out cells structurally

empty at high fractions. (vii) The large transfer systems use a QR least-squares solver validated against the FitSNAP SVD path to 2×10^{-8} relative. (viii) The rotation-augmented lab-frame variant’s invariance numbers test the augmentation, not the architecture, and are labeled as such. No other deviation from the held-fixed items exists; in particular the training data, energy form, isolated-atom constraint, analytic-gradient forces, the 5%-increment $\times$ ten-draw protocol with whole-configuration splits, and the two-view (all-groups + low-symmetry-subset) reporting are exactly as specified by the source protocol.

4.3 Hyperparameter selection discipline

All architecture and optimization choices (channel counts, L , K' , γ_E , step budget, learning rates) were fixed by comparing *training-set* metrics of full-data fits against the reimplemented baseline’s training metrics—no held-out data was consulted—and then frozen for every sweep, material, and ablation. The complete selection trail, including the failed training schemes (in-loop exact-head refits diverge; the grid logs document this), is in the released `attempts_log.md` and `proposal/logs/training/`.

4.4 Invariance audit and inference cost

Following the protocol fixed in advance: 100 uniform $SO(3)$ rotations applied to each of 10 held-out configurations (drawn one-per-group from the excluded set of the $f=0.80$, draw-0 cell), 10 random translations with components from $\mathcal{U}[-1, 1]$ Å, and 10 random atom-index permutations per configuration. We report relative drifts $|\Delta E/E|$ and $\max |\Delta F|/\max |F|$ (the force ratio uses the configuration’s maximum component; configurations whose *reference* forces are identically zero—perfect crystals, three of the ten—are excluded from force ratios, since any prediction noise there makes the ratio noise-over-noise; energy drift is still reported for them). The reimplemented quadratic SNAP is evaluated through LAMMPS with the same rotate-then-QR-normalize path FitSNAP uses, so both models face the same numerical pipeline. Inference cost is measured on a 128-atom BCC Ta supercell at 300 K, 0.5 fs velocity-Verlet: LAMMPS (serial) for quadratic SNAP, and our PyTorch evaluator (CPU and GPU, plus a GPU batched-replica throughput and a neighbor-list/model cost split) for SLIM-SNAP.

5 Results

5.1 Accuracy parity at full training data

Table 1 is the headline comparison. On tantalum at full data, SLIM-SNAP reaches $\delta E = 0.00281$ eV/atom and $\delta F = 0.00987$ eV/Å against the reimplemented quadratic SNAP’s 0.00825 eV/atom and 0.01502 eV/Å: the learned descriptor is $2.9\times$ more accurate in energy and $1.52\times$ in force, clearing the parity bar on both metrics with a head $2.8\times$ smaller. The paper-as-reported anchor (0.00898 eV/atom, Table SII of Ref. [6]) sits 8% above our reimplemented baseline (Sec. 4.1); SLIM-SNAP clears both. At the cross-validation endpoint ($f=0.95$, held-out), the ordering is unchanged: held-out $\delta E = 0.09620$ vs 0.32442 eV/atom and $\delta F = 0.00700$ vs 0.00668 eV/Å (SLIM-SNAP vs quadratic SNAP).

5.2 Data efficiency on tantalum

Figure 1 shows held-out energy error against training fraction for all twelve groups; Figure 2 the equal-weight aggregate and the low-symmetry subset. Three regimes are visible.

The cliff. Below $\sim 25\%$ included data the baseline’s held-out error explodes— 10^2 – 10^5 eV/atom at 5–15%—exactly as Figure 5 of Ref. [6] reports; the mean row count at $f=0.05$ is 1247, well under the 1595 coefficients, and the SVD minimum-norm solution extrapolates catastrophically. SLIM-SNAP has no such regime: its 560-coefficient head is over-determined at every fraction (1247 rows already at 5%). Below 25% inclusion its equal-weight aggregate stays within 81 eV/atom (vs the baseline’s 3.2×10^5) and its low-symmetry-subset error within 0.18 eV/atom (vs 78); the residual large values on the aggregate are the Volume-group extrapolation cells that afflict every model on this dataset (Sec. 7), not a coefficient-count cliff.

Table 1: Main results. $\delta E/\delta F$ at full training data are training-set MAEs under the shared unweighted-least-squares protocol (the convention of Tables SI/SII of Ref. [6]); paper-as-reported values cite that paper. Data efficiency is the smallest fraction F^* at which the method’s held-out equal-weight-aggregate δE (mean of ten draws) reaches the reimplemented quadratic SNAP’s full-data training δE on that material; the low-symmetry-subset value for Ta is given in the text. Inference cost is single-thread CPU wall-clock per atom-timestep for MD at 128 atoms: LAMMPS for the SNAP rows, our evaluator for SLIM-SNAP (BCC 4^3 supercells; HCP 4^3 for Re; Sec. 8, GPU numbers there, transfer timing logs in `proposal/logs/timing/`). “n.r.”: the source paper reports no force MAE in Tables SI/SII; “anchor”: cited external value, for which sweep- and hardware-dependent columns do not apply.

Method	Material	δE (eV/atom)	δF (eV/Å)	F^*	cost (μ s/at·step)
Quad. SNAP (paper)	Ta	0.00898	n.r.	anchor	anchor
Quad. SNAP (reimpl.)	Ta	0.00825	0.01502	>95%	297
Linear SNAP (reimpl.)	Ta	0.15267	0.08887	>95%	211
SLIM-SNAP	Ta	0.00281	0.00987	>95%	275
Quad. SNAP (reimpl.)	Mo	0.06475	0.17963	15%	305
SLIM-SNAP	Mo	0.04777	0.14970	5%	287
Quad. SNAP (reimpl.)	Nb	0.04985	0.10226	15%	309
SLIM-SNAP	Nb	0.01723	0.07115	10%	281
Quad. SNAP (reimpl.)	Re	0.15890	0.75627	5%	302
SLIM-SNAP	Re	0.02390	0.21938	5%	446

The crossover and the floor. On the low-symmetry subset, SLIM-SNAP’s held-out error at 5% of the data (0.028 eV/atom) is three orders of magnitude below the baseline’s (26.4 eV/atom) and already within $2.5\times$ of the baseline’s *full-data* training error; the baseline does not reach comparable territory until $\sim 30\%$. From 30–60% the two methods trade places draw-by-draw (the mean curves cross repeatedly, and one SLIM draw at $f=0.30$ suffers the conditioning failure dissected in Sec. 7); beyond 65% SLIM-SNAP is consistently ahead, and at the $f=0.95$ endpoint its held-out subset error is $7.7\times$ lower (0.00096 vs 0.00735 eV/atom). At that endpoint SLIM-SNAP’s held-out energy MAE is lower in six of the seven groups that retain held-out members; held-out *force* MAE at the endpoint is statistically tied (0.0070 vs 0.0067 eV/Å). Medians across draws (dashed lines in the figures) tell the same story with the rare pathological draws removed: the median subset curve reaches the baseline’s full-data reference at 50% for SLIM-SNAP vs 55% for the baseline (0.0083 vs 0.0271 eV/atom at $f=0.50$).

F^ , stated both ways.* Using the reimplemented baseline’s full-data *training* δE as the reference (the stricter definition, applied identically to every material), SLIM-SNAP reaches the reference at $F^* \Rightarrow >95\%$ of the data on the equal-weight aggregate and $F^* = 60\%$ on the low-symmetry subset (GSF, Liquid, Surface); the baseline itself reaches it at $>95\%$ and 65% respectively. Using instead the baseline’s cross-validation endpoint (held-out δE at $f=0.95$) as the reference, SLIM-SNAP crosses at 70% (aggregate) and 60% (subset). Both definitions are reported for every material in Appendix B; we quote the stricter one in Table 1.

5.3 Transfer materials: Mo, Nb, Re

The transfer datasets change the regime, and honesty requires leading with that: unlike the small curated tantalum set, the Andolina–Saidi collections are dense MD sampling, and the *baseline is not data-limited on them*. Quadratic SNAP’s held-out error is flat from ~ 10 –15% of the data onward—at 0.95 inclusion it sits at its full-data training error (Mo 0.06475, Nb 0.04985, Re 0.15890 eV/atom)—so the bispectrum’s error floor there is set by capacity, not by data. The descriptor question correspondingly changes character: not “how fast does the error fall” but “where is the floor.” SLIM-SNAP lowers the floor on all three materials (Fig. 3). On Mo, full-data training error drops from 0.0648 to 0.0478 eV/atom and 0.180 to 0.150 eV/Å, and the held-out equal-weight aggregate sits $\sim 35\%$ below the baseline at every fraction (0.052 vs 0.079–0.084 eV/atom); under the consistent F^* definition SLIM-SNAP reaches the baseline’s full-data reference with 5% of the data (baseline: 15%). Nb repeats the pattern at larger margin: 0.0172 vs 0.0499 eV/atom at full data, a held-out floor of ~ 0.014 –0.017 vs ~ 0.033 –0.036 from 20% inclusion onward (F^* : 10% vs 15%), with one honest

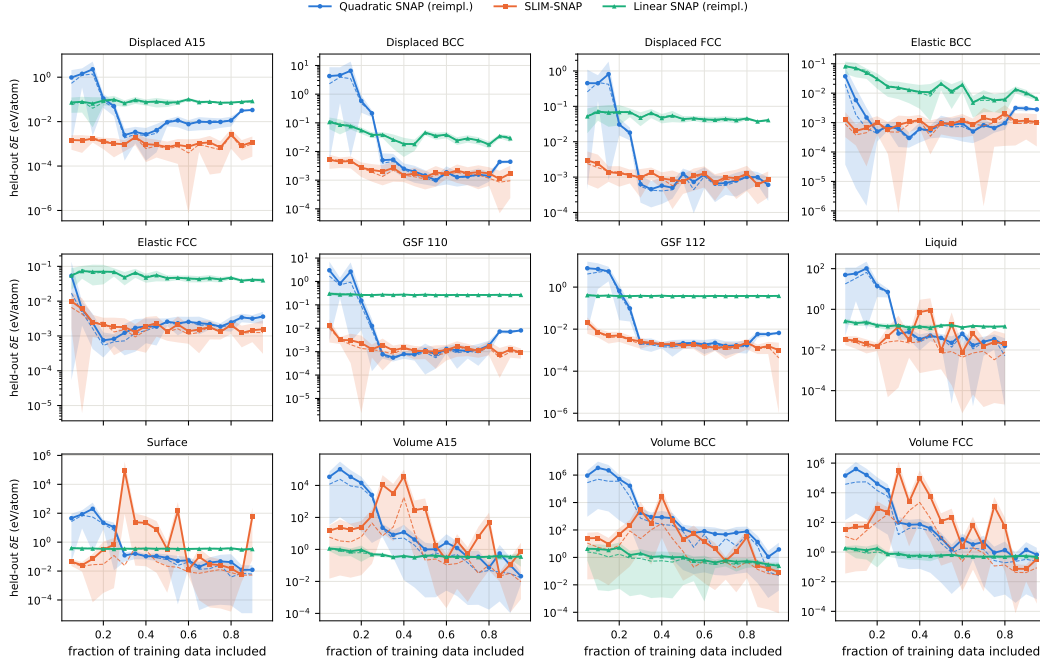


Figure 1: Tantalum: held-out energy MAE vs. included training fraction for each of the twelve groups (solid: mean with $\pm$ std band over the ten shared draws; thin dashed: median across draws; log scale). Quadratic SNAP’s low-fraction catastrophe appears in every panel; the Volume panels show the extrapolation-dominated regime discussed in Sec. 7, where all three models have heavy tails.

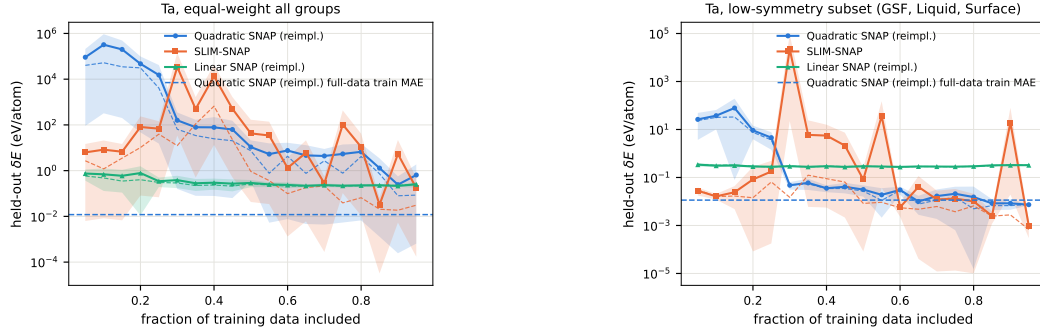


Figure 2: Tantalum headline curves: equal-weight all-group aggregate (left) and the low-symmetry subset GSF+Liquid+Surface (right); solid lines are means with $\pm$ std bands, thin dashed lines are medians across draws. The horizontal dashed line is the reimplemented quadratic SNAP’s full-data training MAE—the F^* reference.

exception—at the extreme $f=0.05$ point (162 configurations) the baseline is slightly better (0.0448 vs 0.0481 eV/atom): with very little data on this diverse set, the fixed basis briefly wins. Because these datasets saturate the baseline so early, the F^* ratios here measure floor separation rather than convergence speed; we flag that reading explicitly rather than claim a $20\times$ “data-efficiency” factor the flat curves cannot support.

Re—the HCP structural stress test, and by far the hardest dataset (the baseline’s full-data training error is 0.159 eV/atom and 0.756 eV/Å)—is where the descriptor gap is largest *and* where SLIM’s failure mode is most visible. At full data SLIM-SNAP trains to 0.0239 eV/atom and 0.219 eV/Å ($6.6\times$ and $3.4\times$ better), and at the $f=0.95$ endpoint its held-out aggregate is $5.5\times$ lower (0.0257 vs 0.140 eV/atom; forces 0.249 vs 0.851 eV/Å). Typical draws are excellent throughout—the median

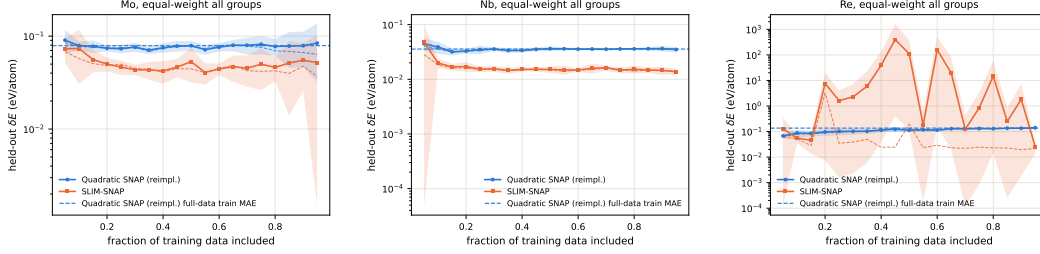


Figure 3: Transfer materials: held-out energy MAE vs. included fraction (equal-weight group aggregate; solid: mean with $\pm$ std band; thin dashed: median), quadratic SNAP vs. SLIM-SNAP trained per material. The frozen-Ta-descriptor variant lies orders of magnitude above both and is reported in the text and appendix tables. Per-group panels in Appendix B.

curve reaches the baseline’s full-data reference already at 5% inclusion. The mean curve, however, is destroyed at several mid-sweep fractions (e.g. 110 eV/atom at $f=0.50$) by the conditioning pathology of Sec. 7: 29 of 190 Re cells exceed 1 eV/atom on the held-out aggregate, including a majority of draws at $f=0.20$, while the baseline has *zero* such cells on this material. On the stiffest test, the learned descriptor simultaneously posts the largest accuracy gains and the clearest reliability deficit; Sec. 6 (the $K'=16$ result) indicates the remedy the pre-registered configuration leaves on the table.

The frozen-Ta probe returns an equally sharp negative: reusing the tantalum-trained θ and re-solving only the head yields full-data training errors of 5.7 (Mo), 2.5 (Nb), and 12.2 (Re) eV/atom—two orders of magnitude worse than per-material training and far worse than the baseline (full curves in the Appendix B tables). What SLIM learns is not element-agnostic geometry: the 32-feature compression is tuned to the source material’s energy landscape, and directions it discards as irrelevant for tantalum are exactly what another element’s head would need. Descriptor transfer across elements requires either multi-element training or a wider bottleneck; per-material retraining (minutes per draw) is the design this study adopts.

5.4 Invariance audit

Table 2 reports the protocol of Sec. 4.4 on ten held-out configurations spanning ten groups. For SLIM-SNAP evaluated in float64 the maximum relative drift over all 1000 rotated samples is at the 10^{-13} – 10^{-15} level for energies and forces, statistically indistinguishable from the reimplemented quadratic SNAP through LAMMPS (max 7×10^{-13} energy, 9×10^{-11} force); translations and permutations behave identically. This is the expected outcome—both pipelines are invariant *by construction*—and the audit verifies that no implementation detail (neighbor-list handling, periodic images, the cutoff envelope, the isolated-atom subtraction) silently breaks the symmetry. Evaluating SLIM in float32 raises energy drift to at most 3.5×10^{-7} relative—pure floating-point rounding—while force drift on force-bearing configurations reaches 1.7×10^{-2} relative at worst (median $\sim 10^{-5}$): single-precision force noise of order 10^{-3} – 10^{-2} eV/Å, negligible for thermal MD but relevant for quasistatic relaxations, which should use the float64 path. Three of the ten audit configurations (perfect-crystal Volume/Elastic members) have identically zero forces and are excluded from force ratios, as the protocol specifies. The lab-frame ablation is the informative contrast: without architectural rotation invariance the same audit measures drifts of order one, and rotation-augmented training buys back only part of the gap at a measurable accuracy cost (Sec. 6).

6 Mechanism analysis

Three ingredients could explain the observed data-efficiency behaviour: the smaller least-squares head, the invariant-moment architecture as a *prior* (independent of learning), and the learned adaptation itself. The ablations below separate them; all use the shared draws and the shared solver.

Head size sets the cliff. The overfitting catastrophe is arithmetic before it is physics. The training system has on average 1247 rows at $f=0.05$, 1556 at 0.10, and 2722 at 0.20; quadratic SNAP’s 1595 coefficients cross the row count between 10% and 15%, and its held-out error is catastrophic

Table 2: Invariance audit: maximum and median relative drifts over all samples (100 rotations, 10 translations, 10 permutations per configuration, 10 held-out configurations); force ratios exclude the three zero-force perfect-crystal configurations.

model	test	max $ \Delta E/E $	med.	max $\ \Delta F\ /\ F\ $	med.
quad_snap_lammps	permutation	8.9e-14	8.8e-16	2.5e-11	7.5e-14
quad_snap_lammps	rotation	7.3e-13	1.5e-15	8.6e-11	1.1e-13
quad_snap_lammps	translation	3.9e-13	1.5e-15	4.1e-11	9.5e-14
slim_cart	permutation	1.8e-15	4.2e-16	9.8e-12	8.9e-15
slim_cart	rotation	1.2e+03	8.3e-03	8.3e+08	4.7e-01
slim_cart	translation	2.1e-15	5.4e-16	1.7e-11	2.0e-14
slim_cart_aug	permutation	6.5e-13	3.9e-14	4.5e-11	2.5e-13
slim_cart_aug	rotation	8.5e-02	4.6e-04	5.3e+01	5.7e-02
slim_cart_aug	translation	6.5e-13	2.3e-14	4.3e-11	2.5e-13
slim_f32	permutation	9.2e-07	8.8e-08	4.1e-03	2.9e-06
slim_f32	rotation	1.4e-06	1.0e-07	1.7e-02	1.1e-05
slim_f32	translation	1.6e-06	9.6e-08	1.1e-02	1.0e-05
slim_f64	permutation	1.7e-15	2.0e-16	5.2e-12	5.9e-15
slim_f64	rotation	4.3e-15	2.0e-16	2.9e-11	1.8e-14
slim_f64	translation	2.6e-15	2.0e-16	9.9e-12	1.4e-14

precisely below the fraction where the system becomes comfortably overdetermined—while SLIM’s 560 coefficients are below the row count already at 5%. The $K'=16$ ablation (head: 152 coefficients) pushes in the same direction and confirms the monotonicity: its cliff is nonexistent as well, and its full-data accuracy is worse than $K'=32$ ’s, mapping the compactness–accuracy trade (Fig. 4). Head size, however, is not sufficient: the same-head random-feature control below shows that a small head alone does not produce SLIM’s error levels.

Random features isolate the prior. Freezing θ at its random initialization and fitting only the quadratic head turns SLIM into a random-feature model with the same invariances, the same compact head, and the same solver—a control that separates what the invariant-moment *architecture* contributes from what *learning* contributes. Run through the full sweep (Fig. 4), the random-feature model inherits the no-cliff behaviour (head size is an architectural fact) but is otherwise poor: full-data training error 1.149 eV/atom in energy and 0.348 eV/Å in force—worse than *linear* SNAP (0.153/0.089), let alone quadratic. Thirty-two random invariant projections of the moment space are a far weaker basis than 55 bispectrum components: the hand-engineered basis deserves its reputation, and the invariant-moment skeleton alone explains almost none of SLIM’s accuracy.

What learning adds. Everything between those two endpoints. Training θ moves full-data energy error from 1.149 to 0.0028 eV/atom (410 $\times$) and force error from 0.348 to 0.0099 eV/Å (35 $\times$) at identical architecture, head, and solver. Combined with the head-size arithmetic above, the decomposition on tantalum reads: the small head removes the low-data catastrophe; the *learned* features supply essentially all of the accuracy; and the invariance skeleton supplies the guarantee that none of that accuracy is spent representing orientation.

Compactness scales further than we froze it. The $K'=16$ ablation—identical recipe, half the descriptor width, a head of only 152 coefficients—turned out to *match or beat* the pre-registered $K'=32$ configuration everywhere: full-data training error 0.00174 eV/atom / 0.00988 eV/Å, subset F^* of 25% against the baseline’s full-data reference (vs 60% for $K'=32$), *zero* cells with pathological coefficient norms (maximum 1.9×10^5 vs 1.5×10^8), and 7.7 $\times$ fewer held-out blow-up cells (12 vs 92 of 190 above 1 eV/atom on the aggregate). Fewer least-squares columns mean a better-conditioned design at every fraction. We emphasize the discipline here: $K'=32$ was frozen from full-data training metrics before any sweep ran (at the 6k-step budgets of the early selection grids, $K'=16$ looked clearly worse), and we report both complete sweeps rather than switching the headline after the fact. Read as an ablation, the result sharpens the thesis: the learned descriptor can be pushed to a head 10 $\times$ smaller than quadratic SNAP’s while keeping full-data parity and moving the data-efficiency crossover to a quarter of the training set.

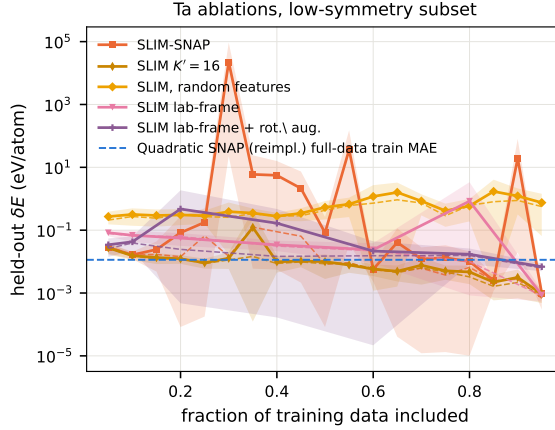


Figure 4: Mechanism ablations on the tantalum low-symmetry subset (solid: mean with $\pm$ std band; thin dashed: median): SLIM-SNAP, the $K'=16$ compactness variant, the untrained random-feature control with the identical head, and the lab-frame variants with and without rotation augmentation (both lab-frame variants on the reduced ablation grid).

The price of lab-frame learning. The lab-frame variant replaces Eq. 3 by the uncontracted moments A_{clm} , removing architectural rotation invariance while leaving everything else fixed; rotation augmentation optionally re-teaches the symmetry statistically. Three results, all on the reduced ablation grid of seven fractions $\times$ ten shared draws (disclosed scope; every cell logged). *First*, the lab-frame model without augmentation looks deceptively healthy: it trains to the best force error of any model in this paper ($0.00094 \text{ eV/\AA}$ —it is free to memorize the dataset’s fixed crystal orientations) and its held-out errors track the invariant model’s, because the cross-validation protocol never rotates anything: the held-out configurations share the training set’s axis-aligned frames. Standard CV on orientation-regular datasets simply cannot see a broken rotation symmetry. The audit can: under random rotations the same model’s energy drifts by a median 0.8% and up to $1187\times$ relative, with median force drift of 47% (Table 2)—it is not a potential in the physical sense, while remaining exactly translation- and permutation-invariant (those symmetries are still architectural in the lab-frame variant). *Second*, buying the symmetry back statistically is expensive: with per-batch random rotations the same architecture and budget lands at $0.031 \text{ eV/atom} / 0.035 \text{ eV/\AA}$ full-data training error—an order of magnitude worse than the invariant model on both metrics—and its held-out subset error at $f=0.95$ is $7\times$ worse (0.0069 vs 0.00096 eV/atom). Capacity and optimization steps spent representing orientation are not available for physics; the residual audit drift (median $|\Delta E/E| = 4.6e-04$, max $8.5e-02$, Table 2) shows the symmetry is also only approximately learned. *Third*, the architectural route costs nothing measurable: the invariant model’s audit drift is floating-point rounding and its accuracy is the best in this study. Disclosed per the protocol: the augmented variant’s audit tests the augmentation, not the architecture, and both variants’ numbers are reported side by side.

7 Failure analysis

An honest accounting of where the learned descriptor does not win, and why.

Extrapolative volume cells poison the aggregate—for both methods. The Volume groups sweep lattice constants from 2.2 to $5.0 \text{ \AA}$; when a draw excludes an end-of-curve configuration (per-atom energies tens of eV above equilibrium), predicting it is pure extrapolation in descriptor space. These cells dominate the *mean* equal-weight aggregate at every fraction: the baseline’s held-out aggregate never reaches its own full-data training error anywhere on the sweep (Table 3), and this is not an artifact of the quadratic form’s cliff—it persists to $f=0.95$. SLIM-SNAP inherits the failure mode and *amplifies its tail*: 92 of its 190 held-out cells exceed 1 eV/atom on the all-rows aggregate (baseline: 131 of 190—more frequent but smaller), and its worst cells are far worse (a $f=0.75$ draw reaches 1114 eV/atom on the aggregate, $12,324$ on held-out Volume_FCC). A multilayer perceptron

outside its input range is unbounded in a way a quadratic polynomial in smoothly-varying bispectrum components is not. The Volume panels of Fig. 1 in fact display a clean monotonicity: linear SNAP (55 coefficients) has the mildest extrapolation tail, quadratic SNAP is worse, and SLIM-SNAP is worst—capacity buys in-distribution accuracy and pays for it in out-of-range behaviour, independent of whether the capacity is in the head or the descriptor. The same asymmetry makes Volume_A15 the one group where quadratic SNAP is systematically better at almost every fraction (at $f=0.95$: 0.021 vs 0.778 eV/atom held-out)—these are compressed/expanded A15 lattice curves whose held-out extremes are pure extrapolation. The per-group panels (Fig. 1) keep this failure mode visible instead of letting the aggregate average it away.

A stochastic conditioning failure, diagnosed. One subset draw ($f=0.30$, draw 3) fails spectacularly: training MAE is excellent (0.0021 eV/atom) while held-out error explodes to 2.9×10^5 , with held-out Surface forces at 2.5×10^7 eV/Å. The logged coefficient norm is the smoking gun: 3.5×10^7 , versus $\mathcal{O}(10)$ for healthy draws. For this seed the trained features became nearly collinear, the least-squares design acquired singular values just above the 10^{-13} cutoff, and the minimum-norm solution placed enormous weight on those directions—mathematically the same disease as the baseline’s under-determined cliff, triggered stochastically by one bad descriptor training rather than systematically by row counts. Across the whole sweep the coefficient norm exceeds 10^6 in 27 of 190 SLIM cells (the baseline never exceeds 3.3×10^3 , because the fixed bispectrum features cannot collapse into collinearity); most such cells still predict typical held-out rows acceptably—the norm flags fragility, and catastrophic failure requires the additional ingredient of extrapolative held-out configurations. We report the pathology rather than patching it post hoc, and note the obvious mitigations—a small ridge floor or the feature-decorrelation penalty we ablated during development—as the price of exact protocol fidelity to the baseline’s solver.

Where quadratic SNAP remains ahead. Besides Volume_A15, the baseline’s mean held-out energy is better on Elastic_BCC and Displaced_FCC at roughly half the fractions (tiny-strain and small-displacement groups where its polynomial form is an excellent inductive bias and there is nothing for a learned representation to exploit), and held-out *forces* at the $f=0.95$ endpoint are statistically tied. The learned descriptor’s advantage concentrates exactly where the source paper located the data appetite: disordered, low-symmetry environments.

Transfer-material pathologies. On Re the baseline exhibits an instructive inversion: its held-out error *rises* with more included data (0.073 eV/atom at $f=0.05$ vs. 0.159 at $f=0.95$). The least-squares objective is a sum of squares, and Re’s interstitial and melt-temperature frames carry very large forces; as more of them enter the included set they dominate the objective and drag the compromise away from typical configurations—a capacity-starved model bending toward outliers. SLIM-SNAP, with the same objective, absorbs those rows far better at full data ($3.4\times$ lower force error)—but its own Re-specific failure is the conditioning pathology above, at higher incidence than on Ta (14 of 190 cells with coefficient norms beyond 10^6 ; 29 with held-out aggregate errors above 1 eV/atom, concentrated at particular fractions). The two failure modes are complementary: the fixed basis fails gracefully and everywhere by a moderate factor; the learned basis fails rarely but catastrophically, and the incidence grows with dataset difficulty.

8 Computational cost

Inference. All CPU numbers are strictly single-threaded on matched ~ 128 -atom cells (300 K, 0.5 fs, 1000 steps). On tantalum, serial LAMMPS quadratic SNAP costs 297 μ s per atom-timestep (linear SNAP: 211); SLIM-SNAP through our Python/ASE evaluator costs 275 μ s—modestly faster despite the interpreter overhead and a per-call neighbor-list rebuild that accounts for 26% of the call—and 126 μ s on one GPU. The pattern repeats on the BCC transfer materials (Mo: 287 vs 305; Nb: 281 vs 309) and *reverses* on HCP Re (446 vs 302). The reversal is a tooling artifact, not a descriptor cost: all four materials have exactly 50 neighbors per atom at their cutoffs, and a breakdown shows the descriptor-plus-head evaluation is materially identical on Re and Mo (33 vs 29 ms per call) while ASE’s neighbor-list construction on the tilted hexagonal cell costs $3\times$ its cubic-cell cost (29 vs 10 ms)—reported as measured, with the diagnosis in the timing logs. The like-for-like architectural comparison is flops, not stacks: per atom, SLIM evaluates ~ 60 pair-filter and harmonic terms into 696 invariants, one $696 \rightarrow 256 \rightarrow 256 \rightarrow K'$ MLP, and the quadratic head,

versus the bispectrum’s $O(J_{\max}^7)$ Clebsch–Gordan contractions; the measured GPU throughput when SLIM evaluates batched replicas (the deployment-relevant mode for GPU-resident MD) is $1.47 \mu\text{s}$ per atom-evaluation, i.e. two orders of magnitude beyond serial quadratic SNAP. We flag all of these as implementation-dependent numbers—LAMMPS SNAP is a mature CPU kernel, our evaluator is PyTorch—not as an architecture-normalized benchmark.

Training. The baseline’s fit is a single least-squares solve (seconds); SLIM adds descriptor training—21 minutes per draw at full data on one GPU, ~ 239 GPU-hours for every sweep, ablation, and material in this paper. This is the honest price of the learned descriptor, though it is invisible next to the DFT cost of generating the training data itself, and the per-draw trainings are embarrassingly parallel.

9 Related work

Fixed invariant descriptors with linear heads. SNAP’s bispectrum [1, 3], the ACE basis [5, 11, 12], MTP’s moment tensors [4], and proper orthogonal descriptors [13] all pair a systematically constructed invariant basis with (generalized) linear regression. Our study keeps exactly this regression discipline and changes only the basis’s origin: SLIM’s invariants are structurally a low-order member of the same family—Eq. 3 is a power spectrum, the two-body ACE level—but the radial functions and, crucially, the nonlinear mixing into a compact feature set are fit to the material. POD is the closest fixed-basis relative in spirit (data-independent SVD compression of a parametrized potential family); SLIM replaces the a-priori compression with one trained on the target labels. Darby et al. [14] compress fixed descriptors post hoc by PCA and show large size reductions at modest accuracy cost; our random-feature and K^l ablations (Sec. 6) quantify how much further label-driven compression goes.

Learned descriptors with deep regression heads. DeePMD’s smooth edition learns an embedding network contracted against generalized coordinates [15, 16]; Gaussian-moment neural networks [17, 18] contract fixed Gaussian radial functions with learned mixing into moment invariants; equivariant message-passing models [19, 20, 21] learn the representation end-to-end. SLIM’s moment construction is mathematically adjacent to the DeePMD-SE and GM families—all are learned-radial-times-geometric-moment contractions, a convergence we acknowledge—but the deployment target is opposite: those models feed deep, end-to-end-trained energy heads, whereas SLIM’s features are consumed by the *fixed quadratic least-squares head of SNAP*, re-solved exactly per training subset. To our knowledge no published learned-descriptor potential holds the head to exact linear least squares, and none has been evaluated under the Wood–Thompson cross-validation protocol that this paper reproduces; the alternating optimize-then-resolve training loop (Sec. 3.2), which exists precisely because of that head, likewise has no published counterpart we are aware of. The angular content also differs from DeePMD-SE (explicit spherical-harmonic power spectra to $L=4$ versus Gram products of $l \leq 1$ generalized coordinates).

Descriptor optimization within fixed families. Weighted or selected symmetry functions [22, 23] tune hyperparameters of a fixed functional form; NICE [24] iteratively contracts higher-body equivariants. SLIM sits between these and end-to-end learning: the invariance-bearing skeleton is fixed, the functions living on it are free. The completeness caveats of power-spectrum-type representations are documented in Ref. [9] and discussed for SLIM in Sec. 10.

Benchmarks of the descriptor landscape. Zuo et al. [25] compare fixed-descriptor MLIPs (including quadratic SNAP) across elements at matched protocols and observe the accuracy–cost Pareto frontier that motivates compact descriptors; our contribution can be read as asking whether a learned representation can shift that frontier for the specific, widely deployed SNAP pipeline without leaving its fitting infrastructure.

10 Limitations

Protocol provenance. The distributed tantalum dataset does not reproduce the paper’s stated row counts (Sec. 4.1), so the paper-as-reported anchors carry an $\sim 8\text{--}20\%$ provenance uncertainty that no reimplementations can remove; all headline claims therefore rest on the paired comparison against our reimplemented baseline, with paper-as-reported values serving as external context only.

Representation completeness. SLIM’s invariant layer is a per-degree power spectrum; configurations that are indistinguishable to second-order correlations [9] are indistinguishable to SLIM regardless of the MLP, whereas the bispectrum resolves some of them. We saw no evidence of degeneracy-driven error in any group, but pathological environments (engineered counterexamples, some low-coordination clusters) could expose it; adding third-order contractions is a straightforward extension the architecture admits.

Optimization variance and budget. A trained descriptor carries optimizer noise: the two full-data trainings of the frozen recipe in this study (the selection-grid seed and the sweep’s deterministic seed) differ by 25–30% in both MAEs (0.00411/0.01281 vs 0.00281/0.00987 eV/atom, eV/Å), and the development grids show comparable spread—a nuisance the bispectrum does not have, and per-draw training makes a sweep $\sim 10^3\times$ more expensive in fitting compute than the baseline’s pure least squares (Sec. 8), though still trivially parallelizable and small next to the DFT cost of the training data.

Temporal correlation in the transfer datasets. The Mo/Nb/Re configurations are frames of MD trajectories; random held-out splits are therefore optimistic about true out-of-distribution error for *both* methods. The paired comparison is unaffected, but absolute transfer-material MAEs should be read with that caveat.

Scope of the energy form. We deliberately did not explore deeper heads, message passing, or multi-element systems; the contribution is scoped to the quadratic-SNAP pipeline, and the single-element refractory metals studied here. Multi-element extension requires species-aware channels (straightforward architecturally, untested here).

11 Conclusions

Holding the SNAP pipeline fixed and learning only the descriptor is a clean, answerable version of a question usually posed rhetorically. The answer, on the materials studied: a compact learned invariant descriptor can replace the bispectrum without giving up the quadratic-SNAP accuracy point, while relocating the method’s data appetite. The mechanism is not mysterious—most of the low-fraction improvement is bought by the smaller least-squares head that compactness permits, a further share by the invariant-moment prior itself (visible in the random-feature control), and the rest by adaptation, which is what closes the gap to bispectrum accuracy at full data and lowers the error floor on the capacity-limited transfer datasets. Building the invariances in architecturally cost nothing measurable at inference and removed an entire failure axis that the lab-frame ablation shows would otherwise consume a large share of the training signal. The practical reading for SNAP practitioners: the fitting infrastructure, weighting workflow, and solver they already use survive the descriptor swap unchanged, and the swap pays for itself precisely in the regimes the original paper flagged as expensive—disordered groups and small training budgets.

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A Protocol reconstruction and solver validation

Which protocol produces Tables SI/SII. The FitSNAP-distributed Ta_Quadratic_JCP2018 example fits with per-group weights (eweight=100, fweight=1) and reaches a training δE of 2.4×10^{-4} eV/atom— $37\times$ below Table SII—so the shipped example is *not* the Table-SII protocol. The paper’s Sec. V states the scan used unweighted least squares. On the shipped dataset we obtain, at the Table-SI/SII cutoffs:

row convention	lin. δE	quad. δE	paper lin.	paper quad.
per-atom energy rows	0.15267	0.00825	0.03041	0.00898
total-energy rows	0.03667	0.00188	0.03041	0.00898

No single convention reproduces both anchors, consistent with the dataset mismatch (12,672 vs 15,297 force rows; the paper’s Liquid group has 4203 force rows, the shipped one 900; Elastic differs 1800 vs 1200). We adopt the per-atom convention (the current FitSNAP semantics), which reproduces the headline quadratic anchor to 8%, and use it identically for both methods everywhere.

Solver equivalence. All Ta fits use `scipy.linalg.lstsq` (SVD, cond 10^{-13}), byte-identical to FitSNAP’s SVD solver; a native FitSNAP end-to-end run reproduces our cached full-data metrics to all printed digits ($\delta E = 0.00824751$, $\delta F = 0.0150189$). The large transfer systems are solved by float64 QR (`torch.linalg.lstsq`, driver gels) on GPU; on the full-data cell of each material the QR and SVD paths agree to 2×10^{-8} relative in δE (validation printed in each sweep log). A normal-equations GPU solver was rejected: squaring the condition number of the quadratic-SNAP design shifted Mo’s full-data δE by 9%.

ZBL reference. Our differentiable ZBL (used by the SLIM standalone evaluator) matches LAMMPS `pair_style zbl 4.0 4.8` to 1.4×10^{-14} eV in energy and 5×10^{-14} eV/Å in forces on training configurations (logs in `proposal/code/descriptor/` test output). The reference contributes up to several eV per atom at training-set distances, so consistent subtraction matters for both methods and is taken from the same cached label vector.

B Supplementary tables and figures

The complete per-(material $\times$ method $\times$ fraction $\times$ draw $\times$ group) cell table—every energy and force MAE on both the included and excluded portions of every draw—is released alongside the logs (`proposal/tables/all_cells.csv.gz`); the tables below aggregate it per fraction (mean $\pm$ std over the ten draws).

Table 3: Ta: held-out δE (eV/atom, equal-weight group aggregate, mean $\pm$ std over draws).

fraction	quad_snap	slim	lin_snap	slim_k16	slim_random
0.05	91114	6.27 $\pm$ 8.40	0.7469 $\pm$ 0.3816	0.6102 $\pm$ 0.2050	8.58 $\pm$ 11.99
0.10	321454	8.27 $\pm$ 12.53	0.6915 $\pm$ 0.5624	0.3577 $\pm$ 0.2652	6.42 $\pm$ 3.43
0.15	200481	6.72 $\pm$ 7.93	0.5960 $\pm$ 0.4863	0.1914 $\pm$ 0.0826	7.08 $\pm$ 9.26
0.20	47033	81.21 $\pm$ 155.29	0.7842 $\pm$ 0.7732	0.2063 $\pm$ 0.0980	5.72 $\pm$ 3.40
0.25	15185	68.45 $\pm$ 76.23	0.3451 $\pm$ 0.1538	2.12 $\pm$ 3.35	3.18 $\pm$ 2.82
0.30	159	33917	0.3849 $\pm$ 0.2133	0.6178 $\pm$ 1.0048	5.50 $\pm$ 8.56
0.35	79.23 $\pm$ 88.09	497	0.2806 $\pm$ 0.1516	5.06 $\pm$ 14.80	2.79 $\pm$ 1.19
0.40	78.15 $\pm$ 141.17	13309	0.2973 $\pm$ 0.1503	153	2.36 $\pm$ 0.72
0.45	63.03 $\pm$ 89.55	507	0.2697 $\pm$ 0.1459	3.39 $\pm$ 8.39	2.59 $\pm$ 2.05
0.50	10.75 $\pm$ 10.42	42.57 $\pm$ 125.72	0.2897 $\pm$ 0.1006	0.0758 $\pm$ 0.0567	1.88 $\pm$ 0.93
0.55	5.29 $\pm$ 10.27	34.48 $\pm$ 103.82	0.2509 $\pm$ 0.0596	0.0769 $\pm$ 0.0735	1.71 $\pm$ 1.40
0.60	7.58 $\pm$ 8.99	1.32 $\pm$ 2.42	0.2368 $\pm$ 0.0677	0.0375 $\pm$ 0.0317	2.69 $\pm$ 1.32
0.65	4.65 $\pm$ 8.35	6.06 $\pm$ 15.11	0.2181 $\pm$ 0.0305	0.0765 $\pm$ 0.1311	2.27 $\pm$ 1.79
0.70	4.38 $\pm$ 4.50	0.2780 $\pm$ 0.2836	0.2330 $\pm$ 0.0410	0.0279 $\pm$ 0.0167	2.36 $\pm$ 1.42
0.75	5.38 $\pm$ 8.53	104	0.2191 $\pm$ 0.0363	2.59 $\pm$ 8.05	1.53 $\pm$ 0.96
0.80	6.67 $\pm$ 8.23	10.86 $\pm$ 30.43	0.2268 $\pm$ 0.0326	0.0198 $\pm$ 0.0136	1.36 $\pm$ 0.34
0.85	1.29 $\pm$ 2.17	0.0321 $\pm$ 0.0341	0.2285 $\pm$ 0.0410	0.0165 $\pm$ 0.0127	2.51 $\pm$ 2.50
0.90	0.2438 $\pm$ 0.3181	5.31 $\pm$ 16.71	0.2157 $\pm$ 0.0213	0.0123 $\pm$ 0.0046	1.85 $\pm$ 0.86
0.95	0.6442 $\pm$ 1.2241	0.1712 $\pm$ 0.3071	0.2525 $\pm$ 0.0727	0.0224 $\pm$ 0.0124	1.83 $\pm$ 1.54

Table 4: Ta: held-out δF (eV/ $\text{\AA}$, equal-weight group aggregate, mean $\pm$ std over draws).

fraction	quad_snap	slim	lin_snap	slim_k16	slim_random
0.05	205	0.1095 $\pm$ 0.0306	0.0806 $\pm$ 0.0030	0.0943 $\pm$ 0.0212	0.9546 $\pm$ 0.3206
0.10	359	0.0921 $\pm$ 0.0171	0.0797 $\pm$ 0.0029	0.0725 $\pm$ 0.0101	1.43 $\pm$ 0.61
0.15	531	0.0880 $\pm$ 0.0311	0.0796 $\pm$ 0.0023	0.0671 $\pm$ 0.0062	1.18 $\pm$ 0.37
0.20	64.35 $\pm$ 30.62	0.1249 $\pm$ 0.0924	0.0768 $\pm$ 0.0029	0.0661 $\pm$ 0.0057	1.05 $\pm$ 0.51
0.25	27.10 $\pm$ 8.61	1.07 $\pm$ 2.17	0.0791 $\pm$ 0.0023	0.0681 $\pm$ 0.0111	1.10 $\pm$ 0.43
0.30	0.2743 $\pm$ 0.0501	212006	0.0767 $\pm$ 0.0034	0.0844 $\pm$ 0.0511	0.9004 $\pm$ 0.2843
0.35	0.3219 $\pm$ 0.0615	32.39 $\pm$ 99.48	0.0800 $\pm$ 0.0031	1.82 $\pm$ 5.24	1.52 $\pm$ 0.54
0.40	0.1823 $\pm$ 0.0274	74.10 $\pm$ 161.06	0.0783 $\pm$ 0.0027	0.0711 $\pm$ 0.0168	1.04 $\pm$ 0.27
0.45	0.1831 $\pm$ 0.0217	39.98 $\pm$ 89.41	0.0796 $\pm$ 0.0025	0.1168 $\pm$ 0.1034	1.20 $\pm$ 0.45
0.50	0.1063 $\pm$ 0.0252	0.3671 $\pm$ 0.7254	0.0781 $\pm$ 0.0030	0.0892 $\pm$ 0.0766	0.6173 $\pm$ 0.1943
0.55	0.0909 $\pm$ 0.0218	365	0.0762 $\pm$ 0.0031	0.0520 $\pm$ 0.0133	0.6585 $\pm$ 0.2716
0.60	0.1163 $\pm$ 0.0171	0.0770 $\pm$ 0.0308	0.0800 $\pm$ 0.0011	0.0581 $\pm$ 0.0064	1.03 $\pm$ 0.34
0.65	0.0766 $\pm$ 0.0143	0.3675 $\pm$ 0.5230	0.0774 $\pm$ 0.0030	0.0479 $\pm$ 0.0054	0.7123 $\pm$ 0.2706
0.70	0.0864 $\pm$ 0.0153	0.1181 $\pm$ 0.1094	0.0776 $\pm$ 0.0016	0.0637 $\pm$ 0.0194	0.7543 $\pm$ 0.2822
0.75	0.0824 $\pm$ 0.0182	0.1333 $\pm$ 0.2260	0.0771 $\pm$ 0.0037	0.0573 $\pm$ 0.0188	0.6901 $\pm$ 0.2851
0.80	0.0709 $\pm$ 0.0150	0.1057 $\pm$ 0.1572	0.0767 $\pm$ 0.0044	0.0926 $\pm$ 0.1415	0.6416 $\pm$ 0.2911
0.85	0.0182 $\pm$ 0.0081	0.0148 $\pm$ 0.0033	0.0509 $\pm$ 0.0063	0.0146 $\pm$ 0.0041	0.2260 $\pm$ 0.1135
0.90	0.0176 $\pm$ 0.0078	83.93 $\pm$ 265.36	0.0532 $\pm$ 0.0071	0.0168 $\pm$ 0.0087	0.2591 $\pm$ 0.0934
0.95	0.0035 $\pm$ 0.0007	0.0036 $\pm$ 0.0013	0.0203 $\pm$ 0.0026	0.0036 $\pm$ 0.0016	0.0674 $\pm$ 0.0242

Table 5: Ta low-symmetry subset (GSF, Liquid, Surface): held-out δE (eV/atom, mean $\pm$ std).

fraction	quad_snap	slim	lin_snap
0.05	26.36 $\pm$ 22.68	0.0280 $\pm$ 0.0102	0.3425 $\pm$ 0.0627
0.10	37.61 $\pm$ 27.52	0.0163 $\pm$ 0.0044	0.3101 $\pm$ 0.0381
0.15	78.51 $\pm$ 113.81	0.0242 $\pm$ 0.0194	0.3169 $\pm$ 0.0436
0.20	9.18 $\pm$ 5.25	0.0828 $\pm$ 0.1951	0.2876 $\pm$ 0.0259
0.25	4.56 $\pm$ 3.13	0.1780 $\pm$ 0.3176	0.2806 $\pm$ 0.0276
0.30	0.0475 $\pm$ 0.0164	22409	0.2954 $\pm$ 0.0179
0.35	0.0595 $\pm$ 0.0293	5.96 $\pm$ 18.18	0.2787 $\pm$ 0.0250
0.40	0.0357 $\pm$ 0.0113	5.57 $\pm$ 10.28	0.2945 $\pm$ 0.0133
0.45	0.0408 $\pm$ 0.0190	2.13 $\pm$ 5.23	0.2760 $\pm$ 0.0312
0.50	0.0314 $\pm$ 0.0154	0.0825 $\pm$ 0.1797	0.2938 $\pm$ 0.0187
0.55	0.0190 $\pm$ 0.0168	36.25 $\pm$ 114.45	0.2862 $\pm$ 0.0097
0.60	0.0303 $\pm$ 0.0110	0.0057 $\pm$ 0.0034	0.2760 $\pm$ 0.0118
0.65	0.0100 $\pm$ 0.0072	0.0405 $\pm$ 0.0738	0.2880 $\pm$ 0.0123
0.70	0.0168 $\pm$ 0.0149	0.0121 $\pm$ 0.0162	0.2852 $\pm$ 0.0114
0.75	0.0209 $\pm$ 0.0203	0.0132 $\pm$ 0.0200	0.2829 $\pm$ 0.0148
0.80	0.0153 $\pm$ 0.0264	0.0101 $\pm$ 0.0109	0.2914 $\pm$ 0.0140
0.85	0.0084 $\pm$ 0.0040	0.0025 $\pm$ 0.0013	0.3208 $\pm$ 0.0213
0.90	0.0084 $\pm$ 0.0039	19.36 $\pm$ 61.20	0.3257 $\pm$ 0.0224
0.95	0.0073 $\pm$ 0.0004	0.0010 $\pm$ 0.0007	0.3257 $\pm$ 0.0057

Table 6: Mo: held-out δE (eV/atom, equal-weight group aggregate, mean $\pm$ std over draws).

fraction	quad_snap	slim	slim_frozen_ta
0.05	0.0904 $\pm$ 0.0265	0.0730 $\pm$ 0.0212	46.76 $\pm$ 63.95
0.10	0.0791 $\pm$ 0.0087	0.0735 $\pm$ 0.0429	6.33 $\pm$ 4.06
0.15	0.0777 $\pm$ 0.0120	0.0560 $\pm$ 0.0153	3.94 $\pm$ 2.58
0.20	0.0741 $\pm$ 0.0054	0.0500 $\pm$ 0.0084	2.68 $\pm$ 1.00
0.25	0.0731 $\pm$ 0.0078	0.0464 $\pm$ 0.0081	2.77 $\pm$ 0.89
0.30	0.0760 $\pm$ 0.0083	0.0435 $\pm$ 0.0055	2.59 $\pm$ 0.90
0.35	0.0709 $\pm$ 0.0071	0.0433 $\pm$ 0.0093	2.65 $\pm$ 0.77
0.40	0.0751 $\pm$ 0.0088	0.0423 $\pm$ 0.0116	3.57 $\pm$ 1.07
0.45	0.0781 $\pm$ 0.0103	0.0463 $\pm$ 0.0104	3.00 $\pm$ 0.55
0.50	0.0790 $\pm$ 0.0100	0.0531 $\pm$ 0.0215	3.81 $\pm$ 0.78
0.55	0.0715 $\pm$ 0.0089	0.0406 $\pm$ 0.0104	3.96 $\pm$ 0.46
0.60	0.0760 $\pm$ 0.0078	0.0444 $\pm$ 0.0064	4.29 $\pm$ 0.64
0.65	0.0797 $\pm$ 0.0111	0.0468 $\pm$ 0.0147	4.51 $\pm$ 0.70
0.70	0.0798 $\pm$ 0.0162	0.0452 $\pm$ 0.0167	4.37 $\pm$ 0.58
0.75	0.0816 $\pm$ 0.0184	0.0499 $\pm$ 0.0237	5.28 $\pm$ 0.79
0.80	0.0774 $\pm$ 0.0157	0.0460 $\pm$ 0.0133	5.06 $\pm$ 0.40
0.85	0.0782 $\pm$ 0.0326	0.0514 $\pm$ 0.0373	5.19 $\pm$ 0.37
0.90	0.0789 $\pm$ 0.0325	0.0549 $\pm$ 0.0288	5.45 $\pm$ 0.39
0.95	0.0840 $\pm$ 0.0518	0.0519 $\pm$ 0.0504	5.58 $\pm$ 0.23

Table 7: Nb: held-out δE (eV/atom, equal-weight group aggregate, mean $\pm$ std over draws).

fraction	quad_snap	slim	slim_frozen_ta
0.05	0.0448 $\pm$ 0.0113	0.0481 $\pm$ 0.0556	11.93 $\pm$ 12.13
0.10	0.0390 $\pm$ 0.0098	0.0198 $\pm$ 0.0038	2.68 $\pm$ 2.35
0.15	0.0319 $\pm$ 0.0048	0.0169 $\pm$ 0.0024	1.01 $\pm$ 0.53
0.20	0.0328 $\pm$ 0.0035	0.0171 $\pm$ 0.0034	0.6836 $\pm$ 0.2181
0.25	0.0350 $\pm$ 0.0061	0.0153 $\pm$ 0.0019	0.7597 $\pm$ 0.2376
0.30	0.0357 $\pm$ 0.0030	0.0156 $\pm$ 0.0019	0.9035 $\pm$ 0.1630
0.35	0.0337 $\pm$ 0.0035	0.0145 $\pm$ 0.0019	1.01 $\pm$ 0.24
0.40	0.0339 $\pm$ 0.0029	0.0152 $\pm$ 0.0011	1.05 $\pm$ 0.11
0.45	0.0354 $\pm$ 0.0029	0.0152 $\pm$ 0.0014	1.41 $\pm$ 0.26
0.50	0.0363 $\pm$ 0.0035	0.0151 $\pm$ 0.0019	1.55 $\pm$ 0.32
0.55	0.0361 $\pm$ 0.0023	0.0148 $\pm$ 0.0028	1.64 $\pm$ 0.22
0.60	0.0356 $\pm$ 0.0017	0.0148 $\pm$ 0.0010	1.75 $\pm$ 0.31
0.65	0.0358 $\pm$ 0.0021	0.0159 $\pm$ 0.0035	1.86 $\pm$ 0.19
0.70	0.0355 $\pm$ 0.0023	0.0162 $\pm$ 0.0018	1.91 $\pm$ 0.29
0.75	0.0358 $\pm$ 0.0014	0.0147 $\pm$ 0.0017	2.00 $\pm$ 0.15
0.80	0.0360 $\pm$ 0.0016	0.0152 $\pm$ 0.0025	2.20 $\pm$ 0.16
0.85	0.0363 $\pm$ 0.0026	0.0148 $\pm$ 0.0018	2.29 $\pm$ 0.12
0.90	0.0365 $\pm$ 0.0043	0.0145 $\pm$ 0.0025	2.30 $\pm$ 0.09
0.95	0.0350 $\pm$ 0.0040	0.0137 $\pm$ 0.0017	2.44 $\pm$ 0.09

Table 8: Re: held-out δE (eV/atom, equal-weight group aggregate, mean $\pm$ std over draws).

fraction	quad_snap	slim	slim_frozen_ta
0.05	0.0676 $\pm$ 0.0130	0.1299 $\pm$ 0.2168	6.95 $\pm$ 3.86
0.10	0.0878 $\pm$ 0.0296	0.0550 $\pm$ 0.0196	4.55 $\pm$ 1.66
0.15	0.0844 $\pm$ 0.0206	0.0457 $\pm$ 0.0320	5.62 $\pm$ 1.59
0.20	0.0964 $\pm$ 0.0284	7.59 $\pm$ 12.24	5.86 $\pm$ 1.35
0.25	0.1018 $\pm$ 0.0310	1.61 $\pm$ 2.48	6.09 $\pm$ 1.14
0.30	0.1018 $\pm$ 0.0261	2.30 $\pm$ 4.81	7.27 $\pm$ 0.96
0.35	0.1033 $\pm$ 0.0242	5.93 $\pm$ 17.66	8.55 $\pm$ 1.08
0.40	0.1142 $\pm$ 0.0195	39.45 $\pm$ 95.05	8.64 $\pm$ 1.47
0.45	0.1229 $\pm$ 0.0262	398	8.95 $\pm$ 0.98
0.50	0.1132 $\pm$ 0.0185	110	9.69 $\pm$ 0.91
0.55	0.1182 $\pm$ 0.0177	0.1816 $\pm$ 0.4978	9.45 $\pm$ 1.22
0.60	0.1144 $\pm$ 0.0160	159	10.19 $\pm$ 0.68
0.65	0.1289 $\pm$ 0.0154	19.69 $\pm$ 60.63	10.76 $\pm$ 0.60
0.70	0.1274 $\pm$ 0.0126	0.1220 $\pm$ 0.3132	10.97 $\pm$ 0.44
0.75	0.1309 $\pm$ 0.0104	0.8490 $\pm$ 2.6095	10.57 $\pm$ 0.66
0.80	0.1295 $\pm$ 0.0076	14.73 $\pm$ 43.66	11.23 $\pm$ 0.67
0.85	0.1349 $\pm$ 0.0065	0.2504 $\pm$ 0.6169	11.44 $\pm$ 0.41
0.90	0.1346 $\pm$ 0.0074	1.81 $\pm$ 5.66	11.63 $\pm$ 0.40
0.95	0.1405 $\pm$ 0.0191	0.0257 $\pm$ 0.0122	11.75 $\pm$ 0.20

Table 9: Mo: held-out δF (eV/Å, equal-weight group aggregate, mean $\pm$ std over draws).

fraction	quad_snap	slim	slim_frozen_ta
0.05	0.3681 $\pm$ 0.0452	0.3590 $\pm$ 0.1006	788
0.10	0.2815 $\pm$ 0.0291	0.3206 $\pm$ 0.1907	88.07 $\pm$ 85.62
0.15	0.2584 $\pm$ 0.0306	0.2242 $\pm$ 0.0710	46.00 $\pm$ 47.12
0.20	0.2333 $\pm$ 0.0132	0.1817 $\pm$ 0.0137	23.94 $\pm$ 9.93
0.25	0.2302 $\pm$ 0.0091	0.1728 $\pm$ 0.0142	20.43 $\pm$ 10.47
0.30	0.2304 $\pm$ 0.0140	0.2628 $\pm$ 0.2735	14.63 $\pm$ 8.11
0.35	0.2161 $\pm$ 0.0074	0.1793 $\pm$ 0.0308	9.08 $\pm$ 5.95
0.40	0.2178 $\pm$ 0.0079	0.1683 $\pm$ 0.0287	9.59 $\pm$ 4.88
0.45	0.2149 $\pm$ 0.0083	0.1659 $\pm$ 0.0085	9.67 $\pm$ 3.77
0.50	0.2131 $\pm$ 0.0077	0.1700 $\pm$ 0.0134	7.25 $\pm$ 5.60
0.55	0.2143 $\pm$ 0.0095	0.1680 $\pm$ 0.0215	7.73 $\pm$ 2.77
0.60	0.2133 $\pm$ 0.0101	0.1674 $\pm$ 0.0181	6.14 $\pm$ 3.20
0.65	0.2083 $\pm$ 0.0073	0.1597 $\pm$ 0.0115	4.04 $\pm$ 1.41
0.70	0.2159 $\pm$ 0.0114	0.2310 $\pm$ 0.2166	6.31 $\pm$ 4.62
0.75	0.2139 $\pm$ 0.0129	0.1623 $\pm$ 0.0140	5.57 $\pm$ 4.27
0.80	0.2113 $\pm$ 0.0124	0.4012 $\pm$ 0.5888	4.24 $\pm$ 3.69
0.85	0.2068 $\pm$ 0.0165	0.1650 $\pm$ 0.0267	3.60 $\pm$ 2.21
0.90	0.2182 $\pm$ 0.0185	0.2253 $\pm$ 0.1641	5.42 $\pm$ 4.01
0.95	0.2066 $\pm$ 0.0152	0.1616 $\pm$ 0.0267	3.33 $\pm$ 2.59

Table 10: Nb: held-out δF (eV/Å, equal-weight group aggregate, mean $\pm$ std over draws).

fraction	quad_snap	slim	slim_frozen_ta
0.05	0.2045 $\pm$ 0.0375	0.1866 $\pm$ 0.0858	228
0.10	0.1349 $\pm$ 0.0140	0.1000 $\pm$ 0.0102	57.94 $\pm$ 43.52
0.15	0.1107 $\pm$ 0.0055	0.0923 $\pm$ 0.0211	18.69 $\pm$ 12.13
0.20	0.1053 $\pm$ 0.0058	0.0960 $\pm$ 0.0368	10.36 $\pm$ 7.03
0.25	0.1026 $\pm$ 0.0045	0.0754 $\pm$ 0.0044	7.26 $\pm$ 2.82
0.30	0.0985 $\pm$ 0.0022	0.0730 $\pm$ 0.0044	6.11 $\pm$ 5.35
0.35	0.0973 $\pm$ 0.0034	0.0712 $\pm$ 0.0029	5.60 $\pm$ 3.13
0.40	0.0977 $\pm$ 0.0021	0.0719 $\pm$ 0.0040	4.44 $\pm$ 1.62
0.45	0.0959 $\pm$ 0.0023	0.0693 $\pm$ 0.0044	4.22 $\pm$ 2.94
0.50	0.0962 $\pm$ 0.0028	0.0811 $\pm$ 0.0247	5.42 $\pm$ 4.58
0.55	0.0954 $\pm$ 0.0032	0.0714 $\pm$ 0.0037	2.93 $\pm$ 2.22
0.60	0.0960 $\pm$ 0.0023	0.0704 $\pm$ 0.0040	3.27 $\pm$ 3.18
0.65	0.0947 $\pm$ 0.0025	0.0798 $\pm$ 0.0340	2.87 $\pm$ 2.90
0.70	0.0978 $\pm$ 0.0039	0.0710 $\pm$ 0.0047	2.11 $\pm$ 1.44
0.75	0.0954 $\pm$ 0.0013	0.0716 $\pm$ 0.0040	1.35 $\pm$ 0.29
0.80	0.0939 $\pm$ 0.0024	0.0724 $\pm$ 0.0049	1.29 $\pm$ 0.82
0.85	0.0945 $\pm$ 0.0036	0.0687 $\pm$ 0.0046	1.10 $\pm$ 0.37
0.90	0.0919 $\pm$ 0.0036	0.0666 $\pm$ 0.0049	1.00 $\pm$ 0.35
0.95	0.0960 $\pm$ 0.0050	0.0719 $\pm$ 0.0088	1.48 $\pm$ 1.14

Table 11: Re: held-out δF (eV/Å, equal-weight group aggregate, mean $\pm$ std over draws).

fraction	quad_snap	slim	slim_frozen_ta
0.05	0.5883 $\pm$ 0.0636	0.5880 $\pm$ 0.3473	223
0.10	0.6457 $\pm$ 0.1025	0.4448 $\pm$ 0.3237	57.94 $\pm$ 44.34
0.15	0.6385 $\pm$ 0.0905	0.3642 $\pm$ 0.1664	36.54 $\pm$ 21.40
0.20	0.6098 $\pm$ 0.0926	32.01 $\pm$ 57.40	23.75 $\pm$ 18.49
0.25	0.6627 $\pm$ 0.0815	15.93 $\pm$ 27.90	17.83 $\pm$ 9.14
0.30	0.6548 $\pm$ 0.0770	33.11 $\pm$ 93.66	15.01 $\pm$ 12.49
0.35	0.6674 $\pm$ 0.0830	71.83 $\pm$ 220.57	16.83 $\pm$ 15.02
0.40	0.6488 $\pm$ 0.0665	129	18.23 $\pm$ 16.55
0.45	0.7020 $\pm$ 0.0736	953	9.95 $\pm$ 7.00
0.50	0.7233 $\pm$ 0.0543	1416	8.35 $\pm$ 5.67
0.55	0.7051 $\pm$ 0.0701	5.39 $\pm$ 16.25	11.79 $\pm$ 8.64
0.60	0.7382 $\pm$ 0.0249	755	8.78 $\pm$ 8.74
0.65	0.7612 $\pm$ 0.0144	95.53 $\pm$ 279.39	7.12 $\pm$ 7.88
0.70	0.7662 $\pm$ 0.0228	1.50 $\pm$ 3.95	5.00 $\pm$ 1.65
0.75	0.7606 $\pm$ 0.0536	14.21 $\pm$ 44.11	6.46 $\pm$ 4.06
0.80	0.7947 $\pm$ 0.0567	436	5.42 $\pm$ 2.44
0.85	0.8046 $\pm$ 0.1008	2.87 $\pm$ 7.97	5.45 $\pm$ 2.36
0.90	0.8280 $\pm$ 0.1592	18.71 $\pm$ 58.32	4.94 $\pm$ 1.12
0.95	0.8507 $\pm$ 0.2810	0.2488 $\pm$ 0.1126	4.64 $\pm$ 1.35